

WEEK 7 REPORT

Student Performance – Data Cleaning & Transformation Using Power BI

1. Introduction :

During Week 7, I worked on **data cleaning and transformation of the Student Performance in Exams dataset using Microsoft Power BI**. The main objective was to prepare the raw dataset for further analysis by inspecting the data, handling inconsistencies, creating new calculated attributes, reshaping the subject-related data, and preparing basic measures.

The complete process helped me convert the raw student performance data into a structured and analysis-ready dataset.

2. Objective :

The main objectives of this week's work were:

- To import the Student Performance dataset into Power BI.
- To inspect and understand the structure of the dataset.
- To identify and handle missing, duplicate, or invalid data.
- To standardize categorical attributes.
- To create Total Score and Average Score columns.
- To classify students based on their performance.
- To create Pass/Fail categories.
- To reshape subject data using unpivoting.
- To create basic DAX measures.
- To validate the transformed dataset before visualization.

3. Importing the Dataset :

I started by opening **Power BI Desktop** and importing the Student Performance in Exams dataset using the **Get Data** option. After selecting the required file, I opened the dataset in **Power Query Editor** using the Transform Data option.

Power Query Editor was used as the main environment for inspecting and preparing the dataset.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	Completed
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed
23	Male	Group D	Some College	Standard	None

4. Exploring and Inspecting the Dataset :

After importing the data, I examined the available rows and columns to understand the structure of the dataset.

I identified the different types of attributes present in the data, particularly:

- Numerical attributes related to student scores.
- Categorical attributes related to student details and background.

I also checked the data types of the columns and inspected the dataset for possible **missing values, errors, and duplicate records**.

5. Data Cleaning :

The next step was to clean the dataset so that it could be used reliably for analysis.

The cleaning process included:

- Checking for duplicate records.
- Handling missing or invalid values where required.
- Correcting incorrect data types.
- Standardizing inconsistent categorical values.
- Renaming columns when necessary to make them clearer.

This step was important because inconsistent or incorrect data could affect the calculations and analysis performed later.

Table: RemoveColumns(#"Added Conditional Column1", {"math score", "reading score", "writing score", "Total Marks", "Average", "Grade", ...})

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed
23	Male	Group D	Some College	Standard	None

Query Settings

PROPERTIES

Name: Student details

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Capitalized Each Word
- Changed Type1
- Cleaned Text
- Trimmed Text
- Added Index
- Renamed Columns
- Reordered Columns
- Added Custom
- Added Custom1
- Changed Type2
- Added Conditional Column
- Added Conditional Column1
- Removed Columns

6. Transforming Categorical Attributes :

I reviewed and standardized the main categorical attributes in the dataset.

The attributes considered were:

- **Gender**
- **Race/Ethnicity**
- **Parental Education**
- **Lunch**
- **Test Preparation Course**

The values in these categories were checked so that they could be consistently used during the later stages of analysis.

7. Creating Calculated Columns :

One of the important parts of the transformation process was creating new columns from the existing subject scores.

The three subject scores used were:

- **Math Score**

- Reading Score
- Writing Score

Total Score

I created a **Total Score** by adding the three subject scores:

$$\text{Total Score} = \text{Math Score} + \text{Reading Score} + \text{Writing Score}$$

Average Score

I then created an **Average Score** using:

$$\text{Average Score} = \text{Total Score} \div 3$$

These calculated columns provide a single value that can be used to understand each student's overall academic performance.

Custom Column

Add a column that is computed from the other columns.

New column name
Total Marks

Custom column formula ⓘ
= [math score]+[reading score]+[writing score]

Available columns
parent level of education
lunch
test preparation course
math score
reading score
writing score
Total Marks
Average

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

Average

Custom column formula ⓘ

= [Total Marks]/3

Available columns

- parent level of education
- lunch
- test preparation course
- math score
- reading score
- writing score
- Total Marks
- Average

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

8. Creating Performance Grade :

To make the numerical performance easier to interpret, I created a **Performance Grade** using conditional logic.

The grades were classified as:

Grade	Performance Level
A	High Performance
B	Good Performance
C	Average Performance
F	Low Performance

This classification converts the students' numerical performance into simple categories that can be used for further analysis.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

Column Name Operator Value Output

If Average is greater than or... ABC 123 40 Then ABC 123 Pass ...

Add Clause

Else ABC 123 Fail

OK Cancel

10. Reshaping Subject Data :

The original dataset contained separate columns for:

- Math Score
- Reading Score
- Writing Score

I used the **Unpivot Columns** operation to restructure these subject-related columns.

After unpivoting, the resulting fields were organized as:

- **Subject**
- **Score**

This transformation allows the three subjects to be analyzed using a common Subject and Score structure instead of treating each subject as a completely separate column.

Queries [4] < X ✓ fx = Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subject"}, {"Value", "Score"}})

	Student ID	Subject	Score
	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%
1	1	math score	72
2	1	reading score	72
3	1	writing score	74
4	2	math score	69
5	2	reading score	90
6	2	writing score	88
7	3	math score	90
8	3	reading score	95
9	3	writing score	93
10	4	math score	47
11	4	reading score	57
12	4	writing score	44
13	5	math score	76
14	5	reading score	78
15	5	writing score	75
16	6	math score	71
17	6	reading score	83
18	6	writing score	78
19	7	math score	88
20	7	reading score	95
21	7	writing score	92
22	8	math score	40
23	8	reading score	43
24	8	writing score	39

11. Preparing the Data Model :

After completing the transformations, I reviewed the resulting table and checked whether the data structure was suitable for analysis.

I also:

- Removed unnecessary columns.
- Reviewed the transformed table.
- Checked the required keys and relationships.
- Ensured that the data was organized properly for further Power BI analysis.

12. Loading the Transformed Data :

Once the transformation process was completed, I selected **Close & Apply** to load the prepared data into Power BI.

After loading the data, I verified that:

- The calculated columns were available.

- The transformed subject data was correct.
- The unpivoted structure was working properly.

13. Creating DAX Measures :

I created basic DAX measures to support the analysis of student performance.

The measures included:

- **Average Math Score**
- **Average Reading Score**
- **Average Writing Score**
- **Overall Average Score**
- **Overall Pass %**

These measures provide summarized performance values that can be used for reporting and visualization.

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22	Female	Group B	Some College	Free/Reduced	Completed
23	Male	Group D	Some College	Standard	None

14. Validating the Results :

After completing the data transformation, I validated the results by comparing the transformed data with the original dataset.

I checked whether:

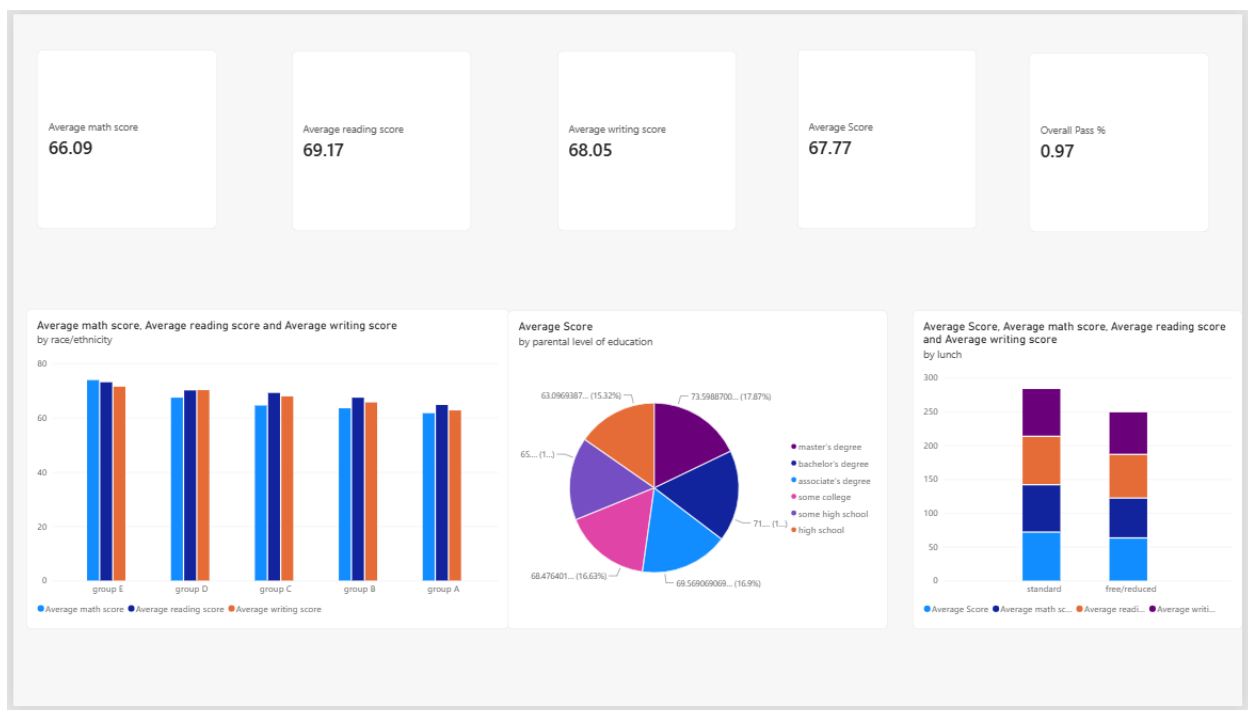
- The calculated scores were correct.
- The Average Score was calculated correctly.
- Performance Grade classifications were working properly.
- Pass/Fail classifications were correct.
- The transformed data was ready for visualization.

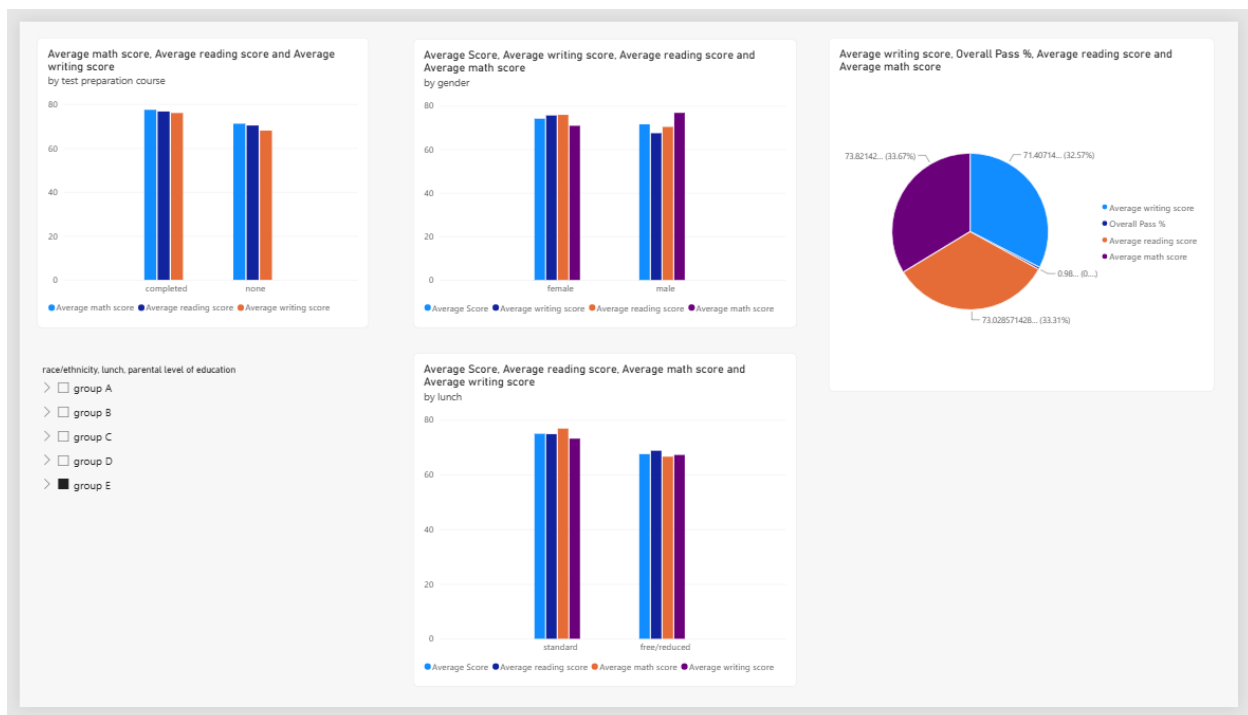
This validation helped ensure that the prepared dataset was reliable before moving to the dashboard development stage.

15. Week 7 Output :

At the end of Week 7, I completed a **cleaned, transformed, and structured Student Performance dataset**. The prepared dataset included calculated attributes, unpivoted subject data, and basic DAX measures.

This prepared dataset became the foundation for the next stage of Power BI analysis.





16. Learning Outcome :

Through this Week 7 activity, I gained practical experience in preparing data for analysis using Power BI. I learned how to work with **Power Query Editor, data cleaning, categorical transformation, calculated columns, conditional classification, unpivoting, data modeling, and basic DAX measures.**

I also understood that proper data preparation is an important step before creating visualizations because accurate and well-structured data produces more reliable analysis.

17. Conclusion :

Week 7 focused on transforming the raw Student Performance dataset into a reliable and analysis-ready format. I completed the process from **data import and inspection to cleaning, transformation, calculated columns, Pass/Fail classification, unpivoting, DAX measures, and validation.**

The final output was a structured dataset containing the required performance attributes and measures, ready to be used for further Power BI analysis and dashboard development.

Submitted By :

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